

35. Answer: D

$$P_{\max} = I^2 R = \left(\sqrt{2} I_{r.m.s.} \right)^2 R = 2 I_{r.m.s.}^2 R$$

Since $I = I_0 \sin(\omega t)$,

$$P \propto I^2 = \{I_0 \sin(\omega t)\}^2$$

$$\therefore P \propto \sin^2(\omega t)$$

Thus the graph should have a waveform in the shape of a $\sin^2(\omega t)$ graph.